

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

End Semester Examination (26/09/2025)

MCA III

Course Name: Cyber Security

Course Code:CS429

Maximum Marks: 25

Total Time: 90 Minutes

Instructions: *Attempt all questions. All parts of a question must be answered in a sequence.*

Q. 1: Short Answer Questions (1 Mark)

- a. Explain what happens during a Denial-of-Service (DoS) attack.
- b. What is the difference between threat and vulnerability?
- c. Differentiate between active and passive attack with examples.
- d. What is the difference between viruses and worms?
- e. Which type of security breach can active attack cause?
- f. What is cyber defamation?
- g. What is Intellectual Property Right (IPR) violation?
- h. What is cyber extortion?
- i. Data diddling is carried out through what?
- j. What does a digital signature ensure?

Q 2. Long Answer Questions (2 Marks)

- a. What is a Trojan Horse? How do they infect the system?
- b. Does software piracy come under intellectual property theft? Explain how?
- c. How would you identify a phishing email?
- d. What are the human failures because of which we become a victim of cyber-attack?
- e. What is CIA? What methods does it use?

Q 3. Answer the questions based on your understanding of case study (1 Mark).

Hario, a 28-year-old illegal alien from Indonesia, has admitted hacking into the business kiosks at US hotel and stealing credit card information that he used to obtain credit. On March 3, 2008, he was sentenced to 10 months in federal prison and ordered to pay \$34,266 in restitution, following his pleading guilty to one count of unauthorized access to a protected computer to conduct fraud. Hario admitted that he hacked into approximately 60 computers operated by Showcase Business Centres Inc. Hario reportedly bypassed four password checks that Showcase Business Centres had in place on their computers, using passwords he obtained while employed by a company that serviced the business kiosks. After hacking into the computers, Hario installed malicious software that allowed him to intercept the data, such as credit card information from customers who used the business kiosks. The malicious software transferred the stolen customer data to a website that Hario controlled, which then enabled him to make fraudulent transactions using stolen credit card data accounts.

- a. Explain how Hario was able to bypass the security mechanisms in place at Showcase Business Centres. What does this reveal about insider threats in cybersecurity?
- b. Which of the following best describes the type of attack Hario carried out?
 - A. Denial of Service (DoS) attack
 - B. Social Engineering attack
 - C. Unauthorized access with malware installation
 - D. Phishing attack
- c. If you were the cybersecurity officer at Showcase Business Centres, what additional security measures would you recommend to prevent future attacks like Hario's? Provide at least two technical and one administrative control.
- d. What role did malicious software play in Hario's attack, and how does data exfiltration via malware typically work in cybercrime cases?
- e. Suppose Showcase Business Centres wanted to prosecute Hario and also prevent reputational damage. What cybersecurity incident response steps should the company have taken immediately after discovering the breach?

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SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

End Semester Examination - 2025

MCA-III Semester

Course Name: Cyber Security

Maximum Marks: 50

Course Code: CS429

Total Time: 180 Minutes

Instructions: Attempt all questions. All parts of a question must be answered in a sequence.

Q 1. Find out the addition, multiplication, additive inverse and multiplicative inverse in modular arithmetic with respect to modulo 9. (5)

Q 2. What is role of Password Manager? Write the features of Gorilla Password Manager. (5)

Q 3. Write RSA algorithm. Find the RSA algorithm d and C for $p = 5$; $q = 11$, $e = 7$; $M = 18$. What are different attack possible on RSA? (5)

Q 4. Sketch the architecture of Chrome Browser and security model. Provide the method for secure browsing. (5)

Q 5. Why does cyber law is necessary component of security? Brief the objectives and sections of chapter eleven of IT Act. (5)

Q 6. Using S-DES, decrypt the string (10100010) using the key (0111111101). The required information are following: (10)

P8								P10								IP									
6	3	7	4	8	5	10	9	3	5	2	7	4	10	1	9	8	6	2	6	3	1	4	8	5	7

IP ⁻¹								E/P								P4			
4	1	3	5	7	2	8	6	4	1	2	3	2	3	4	1	2	4	3	1

$$S_0 = \begin{pmatrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{pmatrix}$$

$$S_1 = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{pmatrix}$$

Q 7. Suppose plaintext for **transposition** scheme is "MASTER OF COMPUTER APPLICATION AT JNU" then what will be the final **ciphertext** for Second order where key is 2 4 3 1 5 7 9 8 6. (5)

Q 8. Why and how do you use the Computer forensic? What are type of Evidence in computer forensic? (5)

Q 9. Describe the KYS and Cyber Persona. What are working of Indian cyber security organization CCTNS and NCCC? (5)